Alex Kiefer ~ Active Inference Insights 014 ~ Representations, Predictive C

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hello everyone and welcome back to
active inference insights today I have
the great pleasure of speaking to Alex
kefir Alex is currently the senior
research and design engineer at versus
uh but was pre previously a lecturer at
uh monach University in philosophy his
work is deeply exciting and unique and
focuses on the convergence of machine
learning computational neuroscience and
philosophy of mind which Alex I'm sure
you know some of my favorite topics the
latter one I feel a little bit more well
versed on but I can't wait to learn more
uh from you about the two prior um
firsty thank you so much for joining me
this is really exciting I think I think
you know reading over your work both
prior to the kind of research that I did
for this and and
during I found it to be incredibly as I
say unique and special because I feel
like you're one of a kind in so far as
not many people are really being really
able to merge the mathematical
side and the philosophical side so I'm
actually just curious more generally
about how you got into this convergence
point was it through maths and uh
machine learning or was it more through
philosophy and then you had to add the
maths on top of
that yeah well first of all thanks so
much for uh for talking with me like I'm
looking forward to just discussing with

you and it's nice that it also happens
to be on a public uh podcast um
absolutely um so so yeah that's a good
question uh so I I came from a
philosophy actually originally came from
an art a fine arts background so I'm I
have I have a tendency to go from one
thing to another but I got into all of
this through philosophy so I would say
as a philosophy gr graduate student I
was thinking about cognitive
architecture you know in a naive way I
was starting to think about these things
uh and this was um you know this was
this was some time ago so I'd say the
landscape was a bit different so not
every philosophy graduate student would
necess necessarily have been exposed to
like deep learning connectionism um
indeed when I was working my on my PhD
deep learning was just starting to sort
of be a thing uh so I started I started
to um find references to like
connectionist models for things like
natural language processing uh and I got
interested in those things um and I'd
say I
mean as is the case for many people in
in my field um Andy Clark's um BBS
article whatever next big water moment
where I i' had been thinking about um
for example um high order basian um
basian models uh signal detection Theory
Of Consciousness for example by people
like hakwan Lao um so I've been thinking
in these terms of generative models
already uh but then that article really
just sort of caused some things to click

into place for me um and I'd had sort of a um I don't know amateur or casual interest in um software development coding on the side and Mathematics um so essentially I got into this through philosophy I picked up as much mathematics as I could off on on the street as it were uh to do the things I wanted to do and I just thought it was really essential to be able to build some of these models to understand really what the hell was going on right so if uh not not to not to get too far into this topic yet but um I mean part of my philosophical point of view is um is that really a naive sort of uh atomistic symbolic approach to content uh the content of mental representations or representations in general is not really um is not really the way to go although it's it's sort of um maybe predominant I'd say in a lot of the philosophical literature so um to really understand something I think you have to understand its functional profile and so I I thought that there were a lot of discussions in philosophy about cognitive architecture that we're not sufficiently grounded in um a kind of knowledge of the the way that these representations might work which I think to some extent I mean there's always um sort of the low road and the high road right there's always top- down understanding and more bottom-up understanding but I think some bottom up understanding um I felt was necessary to

to understand this this stuff and then
um of course once you start to work with
models like this they um sort of become
interesting to work on their in their
own right so to some extent I was pulled
away from the more purely theoretical
questions into wanting to build better
models of cognition sure well I mean
it's in the most praise uh you know
praising way possible that you wouldn't
be able to tell hence why I had to ask
the question um I mean again I'm not a
mathematician so I can't you know I
can't say you know I can't be an
authority uh on your mathematical chops
but honestly it's um I've always I've
always sort of considered you a
philosopher and then I you know reread
the papers I was like wow some of this
is like super technical uh I was really
really impressed so no I find it very I
find it very admirable when philosophers
because I guess it's a path that I'm AI
you know to be candid also trying to
pursue which is supplementing or
combining theoretical thinking with um
you know with mathematical modeling or
mathematical formalisms so it's it's
kind of nice to see you and people like
Maxwell pave that pave that way so
thanks in a weird kind of indirect way
um I'm so glad you brought up content
and representation because that's
actually exactly where I wanted to start
um I kind of thought we would start
somewhat broadly
so I guess um it's funny you know doing
a psychology Masters you probably hear

the word representation quite a lot and
I did uh but no one ever takes no one
ever really goes deeper than that it's
like okay there are representations and
we all kind of have a hunch about what
that means and we think about paintings
or photographs and they stand for
something however just doing this
podcast and doing more reading you kind
of see that it's actually an incredibly
contentious issue what a representation
is do we have representations in the
brain mind uh and so on I was I I feel
like i' come across Notions of um
similarity and resemblance in the
representational literature but maybe
not as sort of directly explicated by
yourself you sort of speak about a form
of basian structural representationism I
thought this was a really really
interesting idea because as you know and
as you
recognize in philosophy resemblance or
similarity and representation are often
divorced um people say well you can't
get representation out of resemblance
something akin to that how do you manage
to get representationalism out of sort
of structural
similarity uh and you know what do your
what does your position entail more
broadly yeah um so so indeed I think
representation is something that it's
it's a term that a lot of people across
many fields use and you know I think
they're justified in using it it's it's
a it's a fairly easy language game to
pick up in a way to start talking about

representations um and I think I think
the term has just has just proven its
usefulness just in the the fruitfulness
of all the various um research programs
that have been that that employ this as
a central term um so I guess I guess to
start with your direct question um I
think one one so right so how do you get
how do you get past the normal the usual
objections to basing um an account of
representation a philosophical Theory
representation and resemblance um I
guess the I guess the the maybe the
first objection that you might encounter
is well resemblances all over the place
um it's very cheap um whereas
representation especially in the sense
that's relevant to psychological um
capacities and theorizing about those
capacities should be something more
special so I'd say the first move to
make is to say that representation is
also cheap um that is to argue that so I
actually have a I'll probably say this a
lot I I say it in any case enough in
conversations I have a paper on this I
haven't
published
um sort of against inflationary concepts
of um mental representation so I think
there are some D you know not dangerous
there are some Waters you get into here
um flirting with things like pan
psychism that maybe we can touch on at
some point but um I think I think here
here's my view if if you can come up
with a serviceable notion of refer
presentation that plays the that that

does the explanatory work it should both
with respect to cognitive science and
also also in common sense terms with
respect to like uh capturing the the
Manifest image as well as the scientific
image right if you can sort of
understand how this this thing that
you're calling representation does those
jobs and if it turns out that it's
everywhere well then it's everywhere
right so I don't think there's any need
to frontload the assumption that
representation needs to be rare um so
that's that's what now I don't want to
say that every resemblance relation is a
representation relation but I think you
can part of the answer to to that
objection is to bite the bullet and say
you know individual neurons probably
represent things for
example uh in in the same sense in the
same fundamental sense as you or I do at
the personal level maybe with different
um features within that right within
that category but we still get the extra
step right so we still get the extra
step from purism resemblance to in your
in your work at Le I mean some people
would just say there are you know at
least in basic Minds there are no
representations and here I'm thinking
about the sort of radical and activist
who say it's only present in these kind
of scaffolded non-basic minds or or
basic forms of mind uh but you do want
the positive representations as you said
and and you do make it clear that we
have to go beyond resemblance so maybe

what I'm kind of leading you here is is towards the kind of Teo semantics or a kind of more functional semantics Tex I really enjoy I I mean I personally really really like this idea um that in some ways a representation means something to the organism I guess because it for me it feeds into ideas of um sense making or affordances or whatever you want to call it so how do we go from Pure objective in quote marks res resemblances to a representation that is for or realized within an organism yeah so all right so I'm going to try to spell this out this is difficult to me this is like in a way it's closest to the core the very core philosophical problems that I think remain sort of unresolved or without a uh uncontentious Revolution resolution for good reason because because it relates to basically the way in which the sort of subject relates to the object the Mind relates to the world so this is the caveat this is still an ongoing um I don't feel settled about about some of these questions um what what what yakob uh Hui and I said in that in in our papers on this um is that um following more or less the account that had been in in the literature um is that you need not just structural resemblance in order to have representation but exploitable structural resemblance um but I do want I do want to emphasize that um I and I can to some degree speak speak for yakob since we

collaborated on this but you know he has
his own views um so I'll just speak for
myself for now um I I don't see either
the resemblance relation or the
exploitability relation as necessarily
um in fact relations to concrete
particulars external to the organism uh
let's start with resemblance um so and
this this is uh this is in the papers
that we wrote although it could stand to
be made more explicit um I think um the
the fundamentally the kind of
resemblance I think that the account
rests on is um resemblance between the
structure of your internal model
um and some hypothetical
world and it needn't be the actual World
um so what you represent in the first
instance and and this this sort of um
resonates with some of the things that
um were said early on in the basian
brain as the spasi and brain ideas
started to break across philosophy and
cognitive science about um you know your
representation of the world being
something like an internal simulation or
perception is like veritical
hallucination or online controlled
hallucination MH um so so the
resemblance in the first instance I
would say is between the internal
cognitive structures and a hypothetical
World which may or may not uh match the
actual world and then I would say a
similar thing about exploitability
um exploitability you might try to catch
that catch that out in very obviously
naturalistic terms in ter in terms of

like as you said teleology um what is
the function of this cognitive structure
what does it tend to allow me to achieve
or what's what did it evolve to allow me
to aeve achieve for example um I am
definitely not a Teo semanticist
actually I think um there's been a lot
of good efforts to integrate active
inference and teos semantics in the
literature um I I think we can get into
this if you'd like but I think it's a I
think it's a misstep I don't think you
should have to look uh at least if you
interpret teleology in terms of um
something historical I think it's a
misstep I don't think you need to look
to any kind of evolutionary history or
beyond the current functioning of the
organism in order to say what it's the
contents of its states art that was
already a mouthful I can always go on
but there there's plenty there so I
guess the first thing I want to just
touch on is so yeah you do speak about
the sort of structural resemblance
between and and this is actually as you
say quite at least in my eyes well
established within the literature the
the the structural resemblance between
the Bas the generative model and the
generative process I really like your
addition there it doesn't have to be the
real world because you've got
imagination memory dreams and so on and
I have had conversations with people who
say that I think you know the perception
Action Cycle itself needs to be maybe
buttressed by uh either retrospective or

prospective ways of thinking and that's
actually kind of in the phenomenology as
well right hle distinguishing between
the live moment and then also
retrospection and prospection more
formally so that's a really nice
addition what what is maybe slightly
more contentious is if we fold truth
conditions in uh or satisfaction
conditions and I I feel like again I
don't want to speak for you you can tell
me that if the generative model is meant
to accurately map onto the generative
process in a way that minimizes that KO
Divergence there has to be some element
of Truth tracking or uh we can't be
completely making it up on the spin
because then our models are basically
useless over time
so how much do you then fold not only uh
the the real- time exploitation but also
the fact that there's got there's some
truth relation between uh world and
model yeah so so I think on the
fundamental um again uh if you just want
to characterize what about this thing
makes it a
representation I think it does I I tend
to think in terms of Truth conditions I
actually think truth conditions are a
nice fit with a basian um approach
because what are distributions over like
events right and they you can think of
these as events and and their
descriptions of states of Affairs not
it's not like your distribution is over
dog right it's over there as a dog
so it's it's a nice fit and of course

you can think of
um of
um well there are nice relationships
between logic as usually defined over
propositions with truth values and
probability Theory so I think that all
works out very nicely
um
so the I guess the question is to what I
mean practically speaking your your
model and this is sort of a very free
energy principle type statement I mean
your model your model will not exist if
it doesn't track the truth to some
degree right and actually I know there's
there's there's definitely a push back
in the um pre energy principle and
active inference literature against the
idea that these models have to be
veritical because they're used for
control of course as well um but I think
I think the the clear truth is that the
your you're of course so perfect
veridicality um at least at the sort of
uh highest possible like um resolution
or Fidelity of representation is is not
needed and it's probably would be
counterproductive right you don't want
the map to be as rich as the territory
usually right but I also think you can't
you absolutely need to to track the
truth to some degree and the nice thing
about structural resemblance approaches
is that you can uh you can quantify
various ways of doing it but you can try
to quantify the degree of structural
resemblance and you don't have to talk
about just like truth or falsity this

this is something that can come inrees
sure okay so that's that we're kind of
in agreement there there's there's only
a slight it's not a caveat it's a kind
of red herring I'm sure that we can
skirt it's actually something I raised
with Carl in our first conversation they
haven't raised since because uh you kind
of get you kind of get integrated into
the basian game and then you forget all
of your sort of actually quite useful
naive questions you know I asked him
Wham if there is just like no numon
right wh if there just are no hidden
States if I just take a very kind of
George Barkley idealist position there's
no kind of all there are ideas I mean I
guess you could posit the ideas as being
the latent States and and their
appearance as being the sensory uh you
know input
but
it's it's a kind of classical
philosophical question which is whams if
we're a brain in the vat do we still get
truth conditions out of that is can we
say that you know we've got hierarchies
of observation and what they track even
within a kind of illusory
space yeah I love that I love that
question
so okay I mean there's different layers
of this there's different ways of
thinking about these skeptical scenarios
um I I would say that you still have
truth conditions for for sure to the
extent
that I mean to some to some degree I'm

just like an intuitive internalist or
you know not solipsist right but
internalist I start from the internal
cartisian kind of point of view and um I
do think there's something sort of
astoic you might say about the way we
represent the world like it seems it
seems as though to me uh that I perceive
things as being thus and so there's sort
of a right and the condition for that
being true I I I think that that so that
representation has a truth condition um
whether or not of course whether or not
it's true or could be true whether or
not the structure of the universe is
such as to you know un to be true even
possibly um I do think uh with respect
to any skeptical there might be versions
of skepticism where you say there's
literally just nothing out there and um
I'm not quite sure what to say about
that but I have thought more about um so
David chamers has this great paper
called I think it's called The Matrix as
metaphysics uh where he goes into some
of these issues and
and um I don't know if I'll be able to
fully justify this at the moment but I
know that where I've landed on this
thinking about this stuff in the past is
that um from the point of view of trying
to model the generative process right so
you figure every agent has its own sort
of subjective ontology which is part of
this it's an aspect of this internal
simulation that you develop over time
through sensory feedback um now whatever
the nature of the the numon is um it's

something that um through interacting
with it has allowed you to generate
these structures uh and I think those
structures um those structures I guess
this a form of structural realism right
so the the struct exist um regardless of
whether even if we're in a simulation um
yeah there's a sense in which the
internal representations that I have
there's truth conditions would be
satisfied unless right so so it could be
part of your overall model that you say
and this isn't a computer simulation or
or something right so so these very high
level um sort of beliefs about the kind
of world you live in your metaphysical
beliefs might might uh falsify your um
your overall model or not um but could
be agnostic on those points sure yeah I
mean I think yob actually has a really
nice contribution to in his um how to
entrain your evil demon paper which is
that like the demon is giving us pretty
reliable uh you know pretty reliable
signals whether it's tricking us or not
I mean we we you there's kind of
beautiful sensory motor Loop whereby we
move and what we sense is changed by
what we how we move and how we move is
changed by what we sense and we seem to
have quite good control over our evil
demon I guess like as just um a kind of
budding philosopher you always run that
argument to the ground which is that
maybe there is no you know maybe there
is no simulation there just is pure
awareness right um or there just you or
there just you or well not you but you

know your mind and you're a solipsist
and in that case there is nothing to
track it's not like a computer
simulation you mentioned David Chas he's
wrar this work on virtual reality which
I'm sure you've checked out and you know
he doesn't think there's such he doesn't
think it really matters right like the
the same structures the same
philosophical relations still hold
whether it's uh carbon or silicon or
whatever but there's still this kind of
binary just distinction okay you've got
what is mapping or the generative Pro
the generative model and the generative
process that which is being mapped
and I just think there is at least
philosophically I might be wrong here a
skeptical position where you just go
well maybe there just is the you know
the sense of mapping but there's
actually nothing behind that right like
all there is is the map there is no
territory that I'd say yeah so even the
of thin sort of realism you get out of
this David Chamas perspective might be
sunk if it's like I forget what this
Russell called it but basically the the
the hypothesis that the world sprang
into existence um a second ago right and
all your memories with all your memories
intact yeah buron Russell right did he
believe so Russell yeah yeah yeah so
yeah so I don't have an answer to that
um I don't think they're I in general
though I think
um yeah so so skepticism is something
that indeed you have you have to kind of

struggle with it I guess as an if you
study philosophy in that way but I I I
think to me the the the outcome of that
struggle is a just a
realization that that's the way things
are and uh and that's okay um it's not
really okay right because if you
actually found out that one of these
scenarios were true and you suddenly
woke up and you're a completely
different world okay that would be be
quite traumatic um traumatizing but well
so would be being alone in the universe
you know that would be quite traumatic
as well honestly it's it's yes um in
most moods it's easy to dismiss but if
you're if you take it seriously honestly
that could be one of that's that's one
of the most terrifying prospects I think
but let's acknowledge it rather than um
um trying to dismiss it via a um you
know superficial argument sure sure I
think in part this is we don't have to
go into this but this is in part a
result of philosophical education at
least in the Anglo uh Saxon World which
is they start you basically on
skepticism um they start you on daycard
and then you
read stuff in Act of inference about
perception and they're all saying well
we kind of take direct realism as you
know for granted you're like what like
we like at school we said that was
rubbish um I just wonder and and then
you know you start hearing all the
externalist attacks on internalism so
I'm kind of curious about your

experience openly and it it sounds like
it's a taboo and it really ought not to
be and I don't think it is but openly
dis or promulgating or proposing a
internalist position it seems to me at
least that from the
outside a fairly rare position to occupy
and the kind of hot thing is to be as
externalist as possible albe there is
some push back of course against radical
and activism so I'm just curious about
sort of sociologically how you found
it yeah no absolutely it's
been it's definitely sort of a um
unpopular position within within sort of
this this this well I'd say in general
as you said in general these days um so
exter so right so before I ever got into
the active inference stuff um I I I sort
of was already fighting the tide against
I'd say
externalism to me it's just it's a
perfectly legitimate set of motivations
right that that motivate the
externalists right I want to understand
the role that representation and related
Notions play in like explanations in
cognitive science and that's a sort of
third person sort of operation and uh
you know uh and so you want to like on
on what basis would I posit contents
well I'm trying to explain you know how
this mouse manages to navigate in its
environment or whatever right um and
there's no reason not to use um sort of
external externalist
um externally available evidence in
doing that um so so that's fine that's

about content extion but I I think
um I don't know I think so so that that
movement I I'm not sure exactly where
the roots begin they probably go back as
far as anything but um there were these
moments in I guess the 70s in the 80s
people like Tyler Burge were really
pushing
anti-individualism uh in the philosophy
of psychology um and I think this is
just one of those cases where the sort
of the baby got thrown out with the bath
water or things right this is like you
know a pendulum and I think the pendulum
swung very far in the direction of
externalism but I just I just think that
leaves out um some very very basic
points like the fact that what we're
trying to explain I I would think and
maybe this is too tied to Consciousness
I'm curious what you think about that
but trying to explain an organism's
point of VI on the world so I think you
want to limit yourself to to that and if
you start to legis about what the
organism's point of view on the world is
by um reading in information about what
you know is in their environment but you
don't know how they're parsing it or how
they're processing the information into
cognitive internal structures I think
you're in danger of um um sort of just
biasing your account of what that of
what their point of view by your point
of view and um that's a sort of naive
way of putting it but I think you can
make some of that more more precise I
well yeah I think I yeah I don't know

how much we need to I think we're kind
of we we're singing from the same him
sheet here I think um I mean I think I'm
in quite a privileged position actually
as a relative Layman because I have no
skin in the game right I just see things
as I see it in terms of my
intuitions and I think also starting
from Consciousness and privileging
Consciousness privileging lived
experience at least biases you towards
some form of internalism I mean I had
this conversation with VZ inz poito in
one of the earliest episodes of this um
series and yeah I I'm exactly like you I
mean I read Evan Thompson and I read
verel and I love it I actually think
it's really beautiful and I love this
idea of the sensory motor Loop and Evan
Thompson actually in mind and life talks
about how the mind body problem can be
become a body body problem it's the
lived body versus the living body but
even if you read that chapter on
Consciousness and Evan I'm sure would
say this and I'm sure he does say it in
the actual book you don't get you still
don't get the phenomenal State you don't
get the phenomenal reality right you
still have that distance and this same
thing I say with VZ which is that well
I'm very happy to say that um
affordances are socially structured by
our language by our culture and so on
but like there's still something it is
like for something to afford itself to
me and I just I have a real hard time
saying that way of seeing the world is

is relative it's in some Vicken steinan
way the qualia is a kind of oldfashioned
idea and let's throw it it just doesn't
do the job for me um I've had a bit of a
I had a very small chat with yakob about
this and I think we're both on the same
page which is
that no well actually and to be I just
don't think anyone's going to be a
really true strong Illusionist about the
hard problem and and
about Consciousness now you can get into
your key Frankish and Dan dennet and say
well it's about qualia and postulating
these private States and again I'm not
really well versed enough to understand
that but I think you're Absol Ely right
starting from philosophy of Mind
analytic philosophy of mind and the
tradition makes one very very skeptical
at least of
reductive um a reductive account which I
guess is actually also you know sorry
for rambling it's it's also kind of
weird because then you see some very
strong externalists also push back
against uh reductivism as I was talking
to Julian kin and Julian is very unhappy
with the idea that neural states are you
and Evan says this as well but they
think well actually you can just expand
that notion into your sensory motor
capacities and your affordances with the
environment like no because you're still
missing something like you're missing
phenomenal reality so I'm kind of
curious about what you think about that
like where where is it do we get out of

the problem by saying you know neurons
aren't new but actually if you talk
about your wider
sociocultural uh you know context and
your history we get
you or or do we need to fold in
phenomenality yeah so I think there are
a couple of issues here um so so one
strand of this leads right into some
other work I've done that the
psychophysical identity stuff where so I
think so the the sort of um
representationalist versus an activist
axis is one thing and certainly I fall
on the representationalist end of that
and I have to admit I probably need to
give an activists they do and and try to
really really immerse myself in the core
literature there but but but to me it's
just it's a non-starter I don't see how
you can
explain just perceptual errors and
things like that without appealing to
something that says that the world is
this way even though it's not and and to
me that's already sufficient to be a
representation and and I don't encounter
many super radical and activists who
think that we don't need representations
so once you have representations in play
then that's that's all I've ever I've
ever assumed um so I don't worry about
too much about that but then you're also
talking about sort of the
I guess the issue of whether the
phenomenal can be reduced to the
physical um things like that and so in
in the psychophysical identity paper I

and I'm not sure by the way of course I don't as a basian and a philosopher I don't have settled views quite generally yeah but the view that I explore in that paper and that I do still want to pursue as a serious possibility is that you can have something like like a mind brain identity thesis so to me
that whatever ever it is it has to do justice to phenomenology yeah um so if we even if it's that you're just your neurons which I I don't think that's quite the story I think probably the scientific account coming from the the life sciences is and um and physics and so on is not is not finished yet and I think wherever we end up with that I don't I think it'll it'll make it sound less absurd whatever it is it's not going to be that we're a collection of neurons necessarily it's probably some some form of self-organization um that that has a sort of natural
we can get into that but but essentially um you'll have that
will even if that's true we I I think a satisfying account has to be able to explain um in completely compelling terms um why it how how that could be true right so how could my current experience of the world while looking at my screen right now uh talking to you how how the hell could that be the same thing as a bunch of you know neuronal activity and so on and I don't personally think that applying to my wider ecological niche or anything like that is going to help elucidate that I

think it's it's more like we need to
elucidate the connection between the
physical mechanisms the physical
mechanism is actually um is actually
stacking the deck too much the physical
processes and my conscious experience
and I think there's some hope that we
can do that but it it cuts along very
different lines from the inactivist kind
of solution yeah well again I I just
personally think that that concession is
made at least within a lot of in
activism even sort of um like alvino's
sensory motor theory talks about how
sort of neural signatures will tell you
maybe why you're seeing red rather than
blue or why you're hearing rather than
seeing but it doesn't tell you why
you're hearing doesn't tell you why
you're saying red yeah I I like your
optimism um I don't know if I share it
if about the about the relationship
between the physical and the phenomenal
uh but I have a soft I have a sort of um
soft appreciation let's say for more
esoteric theories of Consciousness
whether that be a kind of conscious
realism Allah Donald Hoffman or maybe
even pan psychism but again I've got no
commitments um we we'll loop back but I
wanted to just quickly touch upon this
notion of Tio
semantics um which was new to me until a
couple of months ago and then I go into
like was reading about Tios semantics
and Tio semiotics and the difference
between those but if we just stick with
Tios semantics the idea being that sort

of
from what Telos purpose biological
function let's say you can get
representation you can get some meaning
and you wanted to push back on that in a
way because of the historicity so the
content um itself is not necessarily
determined by your
background it's interesting because I
kind kind of see I I kind of see it
differently although I don't necessarily
I don't necess have a good way of
articulating it only because we take as
an assumption that the prior preferences
we have in our active inference models
are at least constituted in part by
evolutionary
priers and I presume that my
representation and also coming from the
affordance literature you know my
representation of a table is very
different from an ant representation of
a table because the affordance is the
differ so what do you mean in terms of
siphoning off The evolutionary history
from some form of Tios semantics yeah oh
great so I mean first I should say I
think Tio semantics is an amazing
research program like I've read a lot of
Ruth milikins work which is foundational
to it and it's fantastic work absolutely
foundational to where we are today um
but I think I mean the big picture is
that I just think as okay I I'll answer
the question you asked which is what do
I mean by siphoning off The evolutionary
history but just quickly to motiv
you have bite some Bulls so I'm to bite

the representation everywhere bullet but
one one thing that milin explicitly says
is so if you have you know there's these
thought experiments like I think it's
called Swamp Man or something where some
some creature appears suddenly out of
out of some Quantum fluctuation or
something and is identical to a uh
physically identical to a human being
who would have evolved over millions of
years Etc right to have the same
structure uh um what miligan says is
that basically the the internal
structures of that creature um such as
the heart brain whatever um shouldn't
really even be called that because those
right those terms refer to things that
have a certain function and that
function comes from the selection
history so I guess the idea is well this
thing just appeared you don't really
know what its parts are for you don't
really know what any for and I just
completely reject that intuition I think
it's clear that if you you know you
might have to spend a couple minutes
analyzing it but if you or talking to to
it depend as the case may be but you
know you'll you'll see you'll be able to
figure out what these things are for
it's like well its heart the purpose of
its heart is to pump blood in the sense
that it's doing that now and it's
keeping it alive and I don't think you
need to look further back um but and and
so I guess I guess that does start to
get into what I mean by by separating
these things off so I think if you have

if you have a rich enough account of the
current functional organization of
something that you should able to figure
out there's a sense in which you can
cash out uh
the the purpose
of the structures within that thing
without knowing about the history so
maybe I'm relying here on on an overly
um complacent distinction between like
constitutive and causal um relationships
so like certainly the history is
necessary to understand how this
structure got to exist I just think
that's a different question from like
what is it doing now possibly what it
purpose now certainly what's its content
so I want to add one more remark here
which is even if you had um a nice way
of Cashing Out biological function um
let's say you could even do it in a way
that doesn't appeal the history I still
don't think that that's even the same
thing I don't think that's the what you
want to explain semantic normativity I
just think it's a different kind of
question yes yes and in fact that's
interestingly the very point that hotto
and my make in rejecting Tio semantics
which is that like you how you gain
content out of biological necessity and
I think Jerry Fodor had a similar stance
um yeah I also wonder whether we need to
make the distinction between what I can
say about the swamp creature and what it
is like to be the swamp creature in
terms of maybe it's not necessarily for
me to know the entire evolutionary past

of the swamp creature in some deep
epistemological ways to account or or to
make a rough approximation of what's
going on
but as you say in terms of the actual
ontological status of its internal
States and by that I mean I I I I buy
the thing about heart and lungs because
you can get those by analogy but I guess
the idea of the kind of private mental
States you probably do get some way by
uh accounting for its evolutionary past
right in the same way that I get some
way from distinguishing myself from an
ant
because well we are very different parts
we've and that evolution is some way I'm
not saying entirely but in some way is
guiding that which you know the
affordances which the landscape presents
to us but no I so so if we then go back
to actually what you do say your s
positive account and give a quote here
um I thought this was really great
so um so you've got sort of yeah your
your structural resemblance and how
we're going to get representation out of
that and you say that um this problem
can be solved by requiring that the
structural resemblance can be used or
exploited by a cognitive system in order
to make cognitive functioning
effective um this captures the intuitive
idea that a representation mental or
otherwise serves as a proxy or standing
for what is res uh what is represented
and I
certainly see that in the sense of how

you would then get representations
because in a sense it's it's meaning for
the
organism some evolutionary biologist
might go well that just you know what
makes something is effective is its
evolution you know it's evolutionary
payoffs I'm still struggling at least
from that to see the strong
divorce from
Tio functionalism or or Tio semantics is
it the addition of the kind of Truth
conditions that get us away from a
purely Tio semantic or fun or functional
semantic
semantics well okay so I guess I don't
see a deep distinction between an
account in terms of Truth conditions and
a Tios semantic account necessarily okay
I thought at least um that what people
like milikin were trying to do is is
provide an account of how you get from
um very
naturalistically
um understandable things
like States carrying information about
other states to
uh and things like natural selection to
an account of representational content
and the main ingredient that's missing
there is the possibility of error um and
I think of those is sort of one and the
same right so that the truth condition
of a representation is what would have
to be the case for it to be accurate or
true and uh you know so if you're if
you've provided an account of uh error
then in a sense you've provided an

account of something that can function
as a truth
condition but I think
I I guess I just don't see so I think
you do need some account some some some
account of the possibility of of error
um so we did discuss uh some ways that
you might get error out in that paper
that you're quoting from I think um so
you could have you could have an
accurate generative model uh that is
it's its structure um resembles the
structure of the generative process in
relevant ways um but it could be
misregistered so you could you know you
could infer that you're in some state
that you're not because of either
because of misapplication of the model
or just because hey it's a probabilistic
model and yeah sometimes what's probable
isn't what's actual right right so so
there's that's that's one way of of
explaining it
um in general I think you can also just
look at the broader even just it's a bit
of a cartoon to say completely
synchronously because any process is
somewhat temporally extended so I would
say history matters only to the extent
that you're trying to give an account of
a process that might exist in some time
scale and so from the point of view of a
smaller time scale that might be a
historical matter but I don't think you
need you should for example you
shouldn't have to go deep or at all into
evolutionary history I think to give an
account of my current perceptual ability

um that is to give an account of the fact that I can perceive things and that I have mental states with contents you do have to appeal to Evolution to explain or or it's it's at least a very good theory if you want to explain like well did I get into the the state in which I have this capacity but I just think those are two different questions sure so I don't think I've really I've really answered your question yet but so I think if you just look at the roughly synchronous um you know present day organization of the organism you should be able to tell what's now this is is this is sort of idealizing I don't think you can actually do this practically speaking in in every case right um You you need to have like a if you could take a snapshot of my physical organization right now and it might take months or years analyzing it to figure out what the actual functional profile of all the components of me as a physical system is or are um if you could do that I think you could tell a story about like well what's the what what is the what is the current functional role of this thing say it's a it's a belief that you know uh belief that there's a fence across the street um and you give an account of the content in those terms in terms of like well which aspect of the generative process does this aspect of the generative model map on to um and yet it could still be the case you could independently investigate whether there

is in fact offence across the street and you could say well this representation has that content but he's wrong about that aspect of the world right yeah yeah yeah yeah no you're absolutely right of course you you you certainly make a point of misrepresentation this is um for those wondering this is representation in the prediction error minimum ization framework by yourself and yob um yeah I think another point you make there which is important is that pure functional semantics don't require this structural mapping I think that's important as well that you don't necessarily need this um relationship between something like a generative process and a generative model in traditional Tios semantics um another thing you add that I thought was really interesting I'd love to learn about and sort of peruse
is this idea of content
holism um because that's a kind of additional part to your proposal and you say here the content of one part of the overall representation depends on the structure of the whole and thus cannot be determined without simultaneously fixing the content of the other parts
and you also I should point out in a footnote I believe allow for that to become recursive so that's not you know the the part is not just the part it
might also be the whole for
subcomponents how what what is Con
holism how did you come to that idea and what does it add to your uh structural

representationism right so I recently
was talking uh with some philosophers uh
who are fellow graduates from my um
philosophy PhD program at The commity
Graduate Center and I um I tend to just
like assume that everyone thinks in
terms of um you know content holism and
stuff and they were like no if you're
gonna submit a paper on this stuff you
really have to explain that because like
people
don't so I realized okay that's
something has to be unack so um this is
something that I got I'd say um
primarily from from Quin from the
philosopher WV Quin who was um so my my
PhD supervisor David Rosenthal was
heavily influenced by Quin and sellers
and they're both uh both theorists who
think along these lines I think this is
one of the great divides in like
philosophical discussions about
semantics um you know in the past past
Century I would just say that um I don't
see holism as something that I'm adding
to the package it's not like something
that I'm sort of tacking on to the view
I think pretty much any functional role
type of semantics which I take
structural resemblance theories to be a
species of um is has has to have this
feature so um if your if the if the if
the view is that your generative model
gets its content in virtue of resembling
the generative process um and by the way
when I just to make this clear because
this this also has come up in some
discussions with people like um at least

when I when I think about that uh I
think of the generative model as
something that has Dynamics and itself
unfolds over time right so so you have a
sort of four-dimensional if you will um
thing that resembles a four-dimensional
external thing or right that might
happen to be externally real as well if
you're lucky if your model is
accurate and and if we're not solists
and if yes and if anything exists sorry
yeah go ahead yeah so um
um right so
the um so the structural right so
so if you want to understand how
anything in that model so so obviously
generative models are and generative
processes are complex things um if you
want to understand the content of any
particular mental state say and let's
say it's realized by some pattern of
activity in prefrontal cortex or
whatever you need to understand and how
what role does that play in My overall
model this four-dimensional object and
then what analogous role does something
play in the system I'm representing and
I just to me that seems like from the
ground up it's going to be a holist
account because the content of this
structure really depends on the role it
plays in the
whole right right I guess uh where I'm
where I probably need clarification is
the idea that part part relations as you
say themselves are not resemblance
relations and then folding that into the
footnote which is that the parts can

also function as structural
representations in these hierarchically
organized systems I guess I don't so if
the content is
fixed according to its representation
the structureal representation as a
whole and thus it doesn't get its
content just purely by the part that it
plays but actually well the you know the
individual part that it plays actually
the fact that it's integrated in a
hole and yet it can also constitute a
hole within let's say a smaller Mark of
blanket if you will
um those can we hold both of those up at
the same in the at the same time um I
think so I'm glad you're talking about
the footnotes by the way I love my
footnotes I would have footnotes within
footnotes if I could footnotes are very
important footnote people people ignore
them I love footnotes yeah they're ter
they're terrible for readers but anyway
they're useful for Artis yeah yeah yeah
so I guess I don't see any any um
conflict here so um for examp say some
some cortical mini column has a plays a
particular role with respect to your to
one's entire brain where it clearly
represents I don't know the the the
location or the pose of some object and
my I say represents that I mean the role
that this mini column plays in the
overall system is analogous to the role
that object plays in the relationships
among things in the in the represented
system right so that's the basic
resemblance um then that that mini

column might also have its internal structure um in virtue of which it represents something but I think we're talking about two different levels of description possibly two different in a in a multiscale active inference picture those might be two different agents yep yep and whether I really want to attribute agency to Min columns and cortical Min columns I don't know but like sure the point is it would be this this subsystem that would have its own um structural representationally grounded content would be a distinct thing that would just be a part of this larger system which is me and only in so far as that thing played the role it does in my generative model would it have content would it constitute part of my point of view on the world yes so okay so maybe can articulate a bit better I guess my question here is well one is a kind of more pragmatic question is about like sort of where do we cap off the structured representation as a whole like that you know where do I end in some sense but we can leave we can park that for one second I guess my point is you say structured representation as a whole gets its content via resemblance the same need not be true of its parts these get their content via the overall structural resemblance but the part part relations themselves are not resemblance relations so the let's say the content of the the component part gets it you know gets it gets that content because

of the overall structural resemblance
but my problem here is okay I of course
I can grant you that and that works for
me but the problem would be then
embedding that in a hierarchy where you
say that that part itself constitutes a
holistic structure who
because you say um we do not mean to
rule out that parts may also be
structured and function themselves as
representations right yeah so but then
that has its own holistic
identity yeah because it's it by you
know by which the its component parts
form their identity according to this
model so I guess I'm just I can imagine
there is a way out but I'm guess I'm
slightly confused about how something
can't have its content fixed
unless it is part of a structured hole
and yet is providing a kind of
structured hole for subparts unless
those subparts in some ways are also
contingent on the overall structured
Hall and we just have a hierarchy of
identity or meaning so yeah so so my
current understanding of this I think it
would be great if it would be really
cool if you could cash out the content
so let's say I have a generative model
whatever phally stitutes it some
internal structures I would say um if we
could if we could sort of show what the
internal relationships are between that
model and lower scale models of
individual neurons for example such that
the contents let's let's say experience
look I'm not attributing Consciousness

to neurons necessarily I think we could
we could leave you can divorce
Consciousness from representational
content to some extent even if you're an
internalist right I'm I still think you
could talk about U maybe this is part of
my deep um the way idiosyncrasies of the
way I was trained in philosophy but I'm
comfortable thinking about you know
unconscious thoughts at the personal
level and such so I don't think I think
you can talk about the content of uh in
of my States divorcing it somewhat from
Consciousness
so we we it would be really cool if like
the content of my current thought had
some intrinsic relationship to the
content of the thoughts of the neurons
right but I don't see that that has to
be the case and that's my my answer to
you is just um whatever lower level
structure it is and so yeah again this
structure could play a role in my the
overall structure let's just say it's my
central nervous system that's the
relevant vehicle I don't know that's but
say it's that it would be in virtue of
the this thing would be basically an
atom in a sense with respect to that
level of
description right all that would matter
would be and this this is definitely
consistent with a hierarchical active
inference point of view what would
matter would be its Markoff blanket
right its internal States can't matter
functionally to the system at that level
right because any internal states that

produce the same input output profile
the same interface would would do just
as well um nevertheless that thing might
be a whole whose
Parts um map onto the parts of some
external
things right it's just that would be a
different system that would be like a
system at a lower level and so it the
cont I I guess one one upshot of holism
it's a bit of a of a hermeneutic type of
picture where the content um is sort of
relative to the or it's It's associated
with the system that you're describing
and so the fact that this thing is a
holistic vehicle for Content um isn't
inconsistent with its being just a part
at the higher level because those are
two different um sort of content
descriptions right content sub Alex and
content
sub
uh cortical column or
whatever you seem very unsatisfied with
that no I'm not I'm not I'm not I'm not
unsatisfied it's it's just a broader
point it's kind of difficult to do these
things in real time in conversation you
know you need to sit down and draw
something or have some time think about
it we can park that we can come back to
it we can we can park it it's uh it's a
fascinating idea and I think it's also
really important for active inference
generally um and predictive code
because you can have this kind of very
coar grained idea of a hierarchy whereby
the top Parts contextualized constraint

determine the lower parts but that kind
of once you get into the granular level
and the philosophically rather vague
point which
is you know priority you know
significance hierarchy priority is a
spectrum and at what you know at what
point we can cast these strict divisions
then I think these questions start to
matter which is about these parle
relations um which actually leads me
very nicely into predictive coding um so
yeah predictive coding in some ways
comes along with active inference and
the Basi and brain hypothesis I think
people have started to be a bit more
strict about distinguishing sort of
continuous State space discreet discret
State space and pondy P schema and
predictive processing but predictive
coding is still a very important part of
literature I really really your paper
where I got a complete history lesson in
predictive coding Helm Holtz
machines uh Jeff Hinton I just I thought
it was great um and people hear these
words and terms thr around like Helm
Holt's machine generative model
discriminative model recognition model
and they they probably don't know what
it means uh and in a sense they they're
relying on an authority like you to hold
the meaning
um so I thought we could kind of
unpicked that not only for the audience
but also for myself um because I take
some of these things just as historical
fact so perhaps we can start with the

distinction between a generative model a
discriminative model and a recognition
model sure
um so with the caveat that I am also
this is also for me because refresh own
memory is always a good thing sure um so
I'd say let's start with generative
versus discriminative discriminative
model um so this is within I'm already
assuming that we're talking about
probabilistic models here um so a
generative model is just basically a
joint probability density over some
variables of Interest right um a
discriminative model uh if you were
going to try to cach that out as like a
um in probab in terms of probability
distributions would would be a
conditional distribution right so if you
have some input X um and some label y um
this model would be a model of the
probability of Y given X right and so
you can use this to um you know these
things are typically used in I guess
discriminative types of of tasks like
labeling images or something uh
classification um generative models are
just much nicer and more powerful in
many ways I think there's all sorts of I
I don't need to rehearse all the all the
arguments in favor of them but no don't
you don't
need so in a from a generative
perspective instead of so a
discriminative model of um like used for
image classification might give you yeah
the pro what's the probability of the
label cat given this input image whereas

you could instead set up a generative
model that jointly models um what what's
what's the joint probability of labels X
and images Y and it could be based on
some conditional um some latent variable
um and then you would just invert that
model given you can and it's a nice
thing about paradigms like this and
predictive coding architectures are very
good at this you can you could Supply an
image and invert the model and get and
get predict a label or you could Supply
a label and predict an image so on and
this and this takes us into the
recognition model um people might be
familiar with the idea of a recognition
density but maybe not a recognition
model as the inversion of a generative
model yeah so this is something where I
need to um if I'm too if I play too
loose and fast with this distinction
then people like Carl and Maxwell will
take me to task for it
so this is one this is one thing that
that has come more clearly in focus for
me over the years which is that um so
the hel Holz machine this this was my my
introduction to all this stuff sort of
computationally was was just coding up
one of those and trying it out and um
and and learning about its properties um
and so in that model you have an an
amortized like a a learned sort of
Frozen recognition model which is
basically it's your it's your it's your
um trained model for inverting the
generative model and so by inverting the
generative model of course I just mean

given some input uh you want to infer
some latent state that that represents
the hidden state that would have caused
that input yeah um so uh you know
obviously doing that in general is
intractable using na inference so you
need you want
some you want to use for example
variational inference uh great so a q
distribution which is your approximate
posterior and the recognition model uh
in the hel holes machine is just a
trained amortized
um par parameterization of the
recognition density so like you give it
some input you have a set of bottomup
Weights that are can be that can be
distinct from the generative weights
that just map into the Laten space so
that's your recognition model because
it's used for recognition yes and yeah
and so from a broader predictive coding
cognitive scientific perspective you
might want to say well clearly organisms
like us Implement some form of
recognition model let's on the basian
brain hypothesis right let's say we have
generative models that go from like
things like Concepts to arrays of of
simulated sensory data you need some way
of getting from the actual data you
receive to the concept back to the
concepts um and so you can think of that
as in a in a broad sense of recognition
model the reason I said I have to be
careful here is that you don't need a
amortized mapping in order to do this in
predictive coding architectures you can

just run the thing um and do iterative inference yeah just try to minimize the free energy over all the nodes in the netork work given an input and so you have a recognition process without necessarily having um a pre-trained recognition model right because the home HS machine has its learning happens in this kind of uh so-call wake sleep cycle right um which I think some people have linked to human sleep but I I W go um I guess I guess a point of confusion for me now I'm not a computer scientist a coder Etc so you are going to have to be patient with me um
is the generative model is talking about these probability densities let's say
just over two
variables um now that seems to me that it doesn't necessarily prioritize one variable over the other like a
discriminative model
would which sounds to me that then inverting it doesn't really make sense because you haven't got a starting version so is the generative model when we talk about a generative model in pred uh predictive coding or active inference the probability let's say the
probability of a state or the
probability of an observation given a state let's say that sounds more like that very you know the line and bases formula sounds like given your explanation like a discriminative model yeah so thanks for the opportunity for clar to clarify this so um yeah so in a discriminative model I my Paradigm for

this is like some feed forward neural
network where you've got image layers
input and N layers processing and like
label layer as
output there to try to train a a Holts
machine for example to do that instead I
would have image and label sort of
concatenated or you could think of them
as two distinct modalities whatever
you'd have these on the bottom layer and
you'd build whatever however many layers
you want on top of that um and so when I
talk about inverting a generative model
so I kind of implicitly assume that the
generative model uh has um latent
variables in it and so the latent
variable it's often called like Z or H
or whatever right so you supply your
input you do some inference onto this
latent variable right and that's that's
your that's your right internal
represent that's your that's your
internal States in an active inference
idiom that would represent the external
States the hidden external States and so
so both the things that so instead of a
mapping from the one let's think of
labels and images as two observation
modalities right instead of instead of
constructing a mapping directly from one
observation modality to another in this
kind of model you would have a mapping
jointly from the observation modalities
to this hidden latent variable and
inverting the generative model is
inferring the latent variable given the
observation I see so it's kind of
embedded in a way um it's kind of you're

you're get you're discriminating
something but within that that itself is
overarching a the generative relation
does that make sense yeah yes exactly so
you could do discrimination right um You
can always get these conditional
distributions from The Joint
distribution so this is just a this is a
way of that's that's what's great about
generative models they do more I mean of
course also harder to train they need
you know it's more information that you
need to model but if you have one then
you can do discrimination um by clamping
a label say inferring a latent State and
then generating the other modality cool
excellent
um yes okay perfect so the the hel Holts
machine
is what 1995
1996 and then friston's predictive
coding I think was introduced first in
2005 but there were probably hints of it
earlier than that um what are the
differences between the two um actually
I'm not going to pit Mr Diane over Mr
friston so I'm GNA leave it I'm I'm not
going to try and ask for a comparison
let's just ask for the the differences
between the two because people probably
hear them aligned but don't know how
they're distinguished yeah yeah I mean I
think a lot of the stuff I I think Carl
and others were working on these ideas
and I think a lot 90s were just there's
amazing stuff going
on but
anyway so yeah so one one difference I

already mentioned is that um in the hel
Holtz machine you explicitly are
training a bottomup recognition model
and as you said this is usually well
originally first proposal for this was
wake sleep algorithm where basically you
would you you'd present an input
propagate the activities given the
current weights and then train the
generative weights to be more likely to
generate that then you would generate a
fantasy top down and retrain the Train
the recognition weights to to produce
that when you get an input um and it was
a way of bootstrapping um two models at
once so the the generative model and the
recognition model um later on and I I
don't know as much about this but um in
discussing this with colleagues um it's
come to light that um something that
worked better was to just was to just
always um just always use the bottom up
activities to train both models so this
is called the re reweighted wake sleep
algorithm so that's that's a tangent
don't need to get into that but um the
wak sleep the hhols machine didn't work
all that amazingly it was just an it was
an amazing proof concept I always
thought of it as like a minimal mind
because it all the basic ingredients you
need sort of imagination perception and
so on maybe not action but we can do
that as um but so so that's that's one
big difference whereas in a predictive
coding Network um
you've got some Vector of um current
hidden State activities um you get an

input uh you make a prediction or you
you send you send a signal up um to the
to the hidden layer um this is the
prediction error um and the prediction
error is computed with respect to the
current activity and you you update the
activities so as to minimize the the
energy term right which just which is
just additive across all the nodes um so
you don't there's no recognition model
um you can just use the uh you can just
use the same the generative weights um
you just just use the inverse of those
to to map to the hidden State um and so
you're not training two distinct models
you just train if you're training a
model like this you would just train the
um you would just train the generative
model by minimizing prediction error um
other differences um so in the hel Holtz
machine there isn't really
any there's a certain um abstraction at
work in that architecture which is that
although so I always intuitively thought
of this as as um modeling the fact that
you know we know from from Neuroscience
that the same neural Hardware is used
for perception and Imagination right at
least largely overlap um and so the hel
machine sort of models that because you
use the same units the same artificial
neurons to do the bottom up sweep as
well as the top down sweep um but those
two information flows never interact
right you've got a wake phase and a
sleep phase and when you're running the
model in generative mode you only use
top down information and so the authors

um Hinton and Di and all pointed out
that this is not really ideal as a model
of perception because we know that there
are top down effects during perception
yep so you don't really want just a a
frozen amortized mapping um that
propagates forward you want to use the
topown information during perception and
so predictive coding has that feature so
if you could just clamp a bottom level
feature Vector whatever whatever the
input is and the the um the predictive
coding
algorithm uh will be
integrating information from the priors
like sort of online in order to perform
the bottom up inference in effect right
so you've got you've always got
predictions coming down from the top and
prediction error coming up from the
bottom and these two sources of
information are integrated online during
perception
so it's a much better model in that
respect Maybe maybe we can also feed in
this idea of um
horizontality in predictive coding um I
actually came to this through Andy
Clark's paper as well whatever next
because he makes quite you know makes
quite an explicit point that it's not
just verticality it's horizontality and
almost the horizontality if I if I'm not
wrong is really kind of where the
inference is happening right it's it's
it's kind of the selection of the best
hypothesis so
maybe you could do a better job than me

uh and just explain you know what what
does horizontal horizontality introduce
to this
picture so I don't know if I am sure how
uh Andre Clark used that word I assume
you mean like lateral connections
ofal no no that's it's fun uh so right
so definitely there's
um there's well so whether or not you
have lateral Connections in your model
right there's there's other ways of
achieving that you can have sort of
diagonal influences like you you know
you use the representations of this
layer propagate information upward and
then there's there's a sort of recurrent
feedback loop where this layer gets
modulated again but you in any case you
get this sort
of I don't know if it's always win or
take all but sort of win or take all
Behavior where one of the hypotheses at
this level selected you can definitely
in some of the earliest predictive coder
predictive coding models were literally
um yeah were not even hierarchical
models they were just like um to to
model um I guess efficient information
processing in retinal ganglia cells I
think where you'd have like the
receptive so so you'd have a neuron here
and it's in it's it's output its spiking
activity would be modulated sort of
dampened by what was going on in its
vicinity right yeah yeah yeah you yeah
you made that point I think in the paper
where yes you talk about the kind of
Center uh I'm butchering this yeah Sur

that's like the early 80s I think yeah
yeah so that was one of the first
applications of predictive coding was
instead of Prior information possibly at
a very deep level right from like
whatever evolutionary priors or or
cognitive priors it's just like well we
sort of embody the prior that um this
cell only needs to fire if what it's
doing is is different from what would be
expected given its neighbors yes um yes
so yeah and and so so and certainly and
and one of I think Carl's earliest paper
on this or one of the early ones I'm
familiar with it's it's called something
like a theory of qu responses responses
thank you um also I've memorized them
all not that I've read them but I've
memorized them that's what you gotta do
I I need to um up my game on that
no so so yeah he discusses the role of
lateral connections um and sort it's
it's you know it's it's sort of a
mechanism for competition in a way it's
it's another way of inducing sparsity
right so if you I've around with um
versions of a hel Holtz Mach machine I
don't know what you call this but I just
added some lateral connections to the
helmoltz machine because it doesn't have
those right I mean well you break
certain theoretical guarantees if you do
that for but but anyway um and what
happens is right you get a increasingly
sparse representation because of this
sort of competitive um potentially
Behavior among the units at one level
yeah yeah that makes sense I guess

um I feel like an issue that I've always
had with predictive coding at a very
very general level and this is more
definitely more indicative of my
intellectual limitations than the actual
theories limitations is it's very
difficult to uh be able to say at a
snapshot in time what's going on I think
over a dionic dionic it kind of makes
sense right like I get these bottom you
know you get these bottom up prediction
errors which in some ways lead to uh
model changes at very you know much
higher levels but still those higher
levels constrain the expectations at the
lower levels but I guess it's kind of
like we're it's it's difficult to
picture it as happening all at the same
time because it's prediction error
trickling up but the predictions that
are going to deal with those prediction
errors are already coming down so why
are those prediction errors coming up in
the first place right if they should be
getting suppressed by the things that
are coming down and maybe some people
skirt this problem just by adopting some
kind of linear time sequence in their
minds I understand it's all happening at
the same time but they go oh like this
this level can't deal with it so it goes
up and eventually you get the prediction
and it suppresses it I've always just
found that to be very problematic
because who is saying at which level the
prediction error stops right it just
seems like it's a kind of
Miracle so maybe you can help me or

maybe you can't help me uh is this an
intellectual is this an ual limitation
of my own or is this something that's
come to mind before um yeah so I think I
can help you to some extent um I don't
know that it's an intellectual
limitation I think
it's I don't see any problem I sort of I
tend to think about this stuff sometimes
in a quite visual way I just sort of
Imagine imagine the stuff working and
then also they try it out and code and
it works so it's okay yeah um so I would
like to give you a more discursive
answer though I think
um um so yes indeed if you try to so and
I've I've definitely built models um
there's David heager has a paper also on
it's a similar title to Carl's paper
it's a theory of cortical something uh
where he has a very similar model and
yeah so so in implementing this it's
like well what do you update first first
of all and like how do you write so I
think the solution to all that stuff
part of it is and I thought a bit about
how to how to build predictive coding
architectures is um just do do the
updating a asynchronously which you know
if if you want a little bit of um
background of that it's basically like
each neuron each model neuron would just
be its own sort of process like so we
can go deeper into the the computer
science stuff if you want but like
basically each neuron would be running
in its own sort of processor thread and
it would take inputs and operate them as

it gets them and then produce outputs as it as it has them to produce and all of this would be going on simultaneously so the idea that you need to first get all of the input to a to a layer the prediction error and integrate it with the top down predictions and figure out how much information to pass on in practice these are just not issues right it's so and even without going into the implementation of this stuff in terms of parallel distributed processing you can just think about um the energy function so the energy the free energy function that you're trying to functional that you're trying to minimize um it's it's additive across the nodes so each node computes its free energy so whenever it gets an input whatever its current state happens to be um
right you just you just modulate its output based on that um so I guess and I guess I imagine this happening so it's it's sort of chaotic I suppose
right it's
um you would just basically each each each neuron could locally process its own its own input and produce an output and and then you know if you're simulating this on a Serial system you have to choose when to do what right um but that's not terribly Central I think to the architecture yes I guess I guess the problem there is if every neuron's just doing its own thing and over time you know over time over the sequence of neuronal activity prediction error just gets sliced off right and so you get

this additive minimization of free
energy you can run a kind of
philosophical thought experiment and
just go well what happens if the total
you know amalgamation or accumulation of
all the neurons Rel to this activity
just they don't do they just don't they
just can't get rid of all that
prediction error but with we're
siphoning off learning right so we're
siphoning off uh parametric updating
because we're saying that's happened
asynchronously I'm just left with a
bundle of prediction error and you never
hear in the fre you know you always say
well that's just going to end up get
minimized
so but yeah so I mean so it's your
prediction error is never minimized
right if were minimized minimized in the
sense of like life would be over you
would be dead yeah um so you know
there's always it's an inter question
you bring up like what what if there's
just prediction error that can't be
handled is it's just going to be this
oscillating like you know chaotic system
where the pred prediction ER just
bounces around yes like that's what
would happen like right you you just you
you just shunt around the prediction
error and you'd have sort of um and I've
seen this happen in in models right
where like you just it just doesn't
converge right you just sort of you just
you just can't um there's no it's not a
stable State it's maybe a maybe a um
metast stable state or something right

where where um there there's there's
oscillatory Behavior something um which
Carl would say is is important right on
to to life on larger time scales yes the
Sol flows yeah so everything is always
sort of smooth so it's it's it's
pathological if you ever get a
probability of a one or zero um these
I'd say in real life as well exceptions
like there might be some we can get this
go straight into like um crazy amounts
of mysticism maybe there's some kind of
fire of one of like existence or
something but like when it
so I think I think so actually but yes
in terms of the probabilities that are
represented by your model like it's it's
it's never going to be no prediction
error um think that's a valid point and
I guess once you fold in the the the
learning whether that's parametric
learning or structure learning you know
that prediction era that was so grave at
one point is eventually going to get
resolved one way or another whether
that's through um you know I find
sometimes when I run into this problem
just anecdotally I find myself going
towards a variational free energy for uh
equation just being like there's my you
know there's my safety because through
either action or perception we're
minimizing you know we're going down
those uh descents of of free energy okay
you you've you've s of you've quashed my
anxiety somewhat um one other sticking
point I have with active inference and
uh this really came to light actually a

couple of weeks ago when I spoke to
Ronald sadki who's a neuroscientist at
University of Vienna and he's Wroten
he's written it's not a word he's
written about the amigdala from an
active inference perspective and he does
this really nice thing of distinguishing
between kind of I think it calls them
preference priors you can call them
phenotypic priors prior what you call
them but prior preferences and then also
what I've been calling State priors
right so literally what state do I
expect to be in now right and you can
have that as your D Matrix or your B
Matrix if we're doing pomdp schema and
then also and and that obviously feeds
into perceptual updating just on like
some if we have just like naive Bays I
know it's not happening like that but
let's just say but then you also have
these preference PS right in many ways
as Cole has made very explicit like that
too is shaping your perception right
hence why we have visual
Illusions I just find in the literature
there hasn't mean too much work gone
into distinguishing those and also
talking about that interplay for you
does that just come out in a
hierarchical system that do those things
Converge on being the same thing we just
call them a prior or is it useful to
continue to distinguish them yeah that's
a great question I mean ideally for me
it's the um I guess the former it's it's
that come out to be the same thing um so
if you're doing pomdp modeling of course

yes you'll Define like a c Vector which is your preference and then you you might also have um well you have you'll also have a d a d vector and so on but I think ideally in a hierarchical system it should be that the these are just you just have your priors and to me what the what the phenotypic um distribution is or the the the Deep sort of priors that represent desires and you know ostatic um requirements to me that's just when your your whole hierarchy links into this something that's sort of at the root no this of this graph right so you have like a basic need for food let's say that's not going to change right you can't really rewrite that or learn learn your way out of it um and then what's go what you're modeling at the lower levels could very well be a consist a situation that's not consistent with that prior being satisfied um but they're still just priors right and really what you want is that the priors at that deep level which are constitutive of and defining of the kind of thing you are you want those to smoothly propagate to the periphery for everything to work out well but of course what you're currently representing at the lower levels of of a hierarchy internal hierarchy is going to be the result of variational inference right so I think of I mean there's some talk about this in the later sections of that paper that Maxwell and Ryan Smith and I wrote on beliefs and desires um where basically

the Q distribution is your sort of your
beliefs in in the barebones version of
the picture um yeah you're that's that's
your estimate of the current state of
the world it might be arbitrarily
different in principle from your desires
um but still um yeah and and the prior
the priors are just the the generative
model itself uh The Joint distribution
that defines this kind of thing that you
are um so I think that
said a lot but I think basically the uh
the distinction between two parameters C
and and um D in a hierarchical model
which is a way that you might set some
of these these priors I think that
that's more of a modeling convenience
than anything I'm not sure that everyone
would agree with me on that but it's
it's really interesting uh and it's a
really useful explanation I mean yeah
once you have them in a kind of smooth
hierarchy you you can start to kind of
visually or mathematically see how it
might all sync up
I guess the issue is in folk psychology
we we still there there's an intuition
always that beliefs preferences let's
say uh sorry desires preferences are
distinct from my beliefs about what's
going on in the world right now and I
guess that's held in some sense because
there's a
hierarchy it it is
yeah sorry I wouldn't say that this
eradicates the distinction I I
understand the the the pull towards that
and this and like some of Andy Clark's

uh where he's
um I don't I think he's he's it's not
clear that he wants to preserve this
this sort of classical distinction
between beliefs and right Desires in a
way um so I understand the pull towards
that but I do think even though your
your desires are these priors which are
belief likee in a way because they to me
they're they're really not right so the
the prior the priors
are right okay so the priors are
certainly they have they they have they
represent something something right and
so they have you know if their desires
then in the classical philosophy of Mind
way of putting it they have satisfaction
conditions right which are a lot like
truth conditions it's just that it's a
different propositional attitude so um
you know they're they're in the same the
representation wise they're in the same
format it's a different attitude to the
same kind of content yes and then the
result of posterior inference is a
belief and so the
same structure cognitive or otherwise I
tend I tend to think that there's a very
transparent relationship between
cognitive and physical sort of
implementation structures anyway the
structure can as considered as post
result of posterior inference it's a
belief um but then considered in terms
of its Downstream effects in in a top
down generative model it sort of
functions as a desire so you have things
in the system yeah in yeah I mean I

wonder whe
How yes the hierarchy is a is a really
useful way in which that I guess again
philosophically vague distinction can be
made
concrete and obviously people also talk
about the temporality aspect of the
generative hierarchy in so far as the
higher levels track slower fluctuations
in the external Dynamics and the lower
levels the faster Dynamics it makes one
think whether that that necessarily maps
onto desires preferences and so on only
being at the higher level that IE being
the result of slower fluctuations right
like I had ice cream I've had ice cream
a thousand times I like ice cream versus
a belief that there is ice cream or that
there isn't ice
cream it's an interesting question it's
an interesting question whether beliefs
sorry well desires let's say necessarily
need to originate at the higher level
and whether sort of more classical folk
psychology beliefs a lower level oh can
I jump in on that no no no I wasn't just
smitb please yeah yeah yeah go I really
like so I think I think Andy Clark also
talks about this in one of his recent
papers more recent papers
so there's a sense in which you really
can talk about sort of like false desire
and versus more genuine desire that
really stems from your being in that I
think sort of genuine sorts of desires
are from the deepest levels of the model
but then in cases of like addiction what
gets happens is yes right so so

basically there's a bottomup sort of an attractor sort of manages to establish itself in your system on the basis of bottomup input that creates a a sharp expectation over a particular outcome that might be not at all a good fit with your deeper priors but that just gets self-confirming account of um of addiction and there's a desire there that will in that it motivates your behavior and you will go for the thing but uh it's definitely deeply obviously um it it impoverishes the rest of the model and right oh that's really nice yeah that's really really nice um John Vu is a very you know big hero of mine talks about reciprocal narrowing and reciprocal opening with respect to addiction I think this isn't his direct work I can't remember whose it is but the idea being that as you said addiction is a kind of downward spiral such that and you can just think about in terms of dopamine um such that a singular Source becomes your kind of source of pleasure in the very folk psychology way but also like your singular prior and everything becomes determined by that and maybe well he he you know he will use the word like Agape but you know you have this kind of opening maybe Hy call Alfa right you have this kind of disclosure and that actually opens yourself up to the world and I guess yeah maybe is a bit mystical but I wonder whether it in doing that we're uncovering something which is actually was already there in a way I mean absolutely I think I think the sort of the sure there's a there's a mystical take on that but it's it's also just um absolutely the case that you want you want as much variability in your States as as is consistent with preserving the phenotype and so y that's something that for example we've written that uh ma and Maxwell and and um um ready and others have written in that resilience uh the work on resilience Mark Miller and yeah yeah Mark yes thank you um I've been doing a lot of things so I haven't thought about that paper in a while but uh but yes like and that just comes out of just VAR I think just variational inference even like you just you just want to maximize the the entropy of your distribution subject to the constraints from your model and yeah Ryan who you mentioned earlier he sort of has this idea of um structure learning aiding emotional granularity and that being really useful because you don't get trapped into thinking that you're Furious if you're maybe just a bit frustrated or and and and so on it's all uh complexity minus accuracy at the end of the day um yeah I don't know what I've had this idea and I'm think you may be writing a paper on it but I there's I think a really nice addition to this whole picture is the idea of selective attention um so Beck mitza who I think was in Carl's lab back in the day he wrote a bunch of papers with Carl about attention and he has this 2019 paper about selective attention I've spoken

about on the podcast before and the idea
being that your priors your preferences
constrain your attention and that kind
of make that's very sort of in sync with
traditional um biased competition Theory
right and is actually stuff that I wrote
about during my masters the idea being
that you know what is elevated in terms
of gain control aligns with is actually
in line with your goals so if I'm
looking for a ginger person then all the
templates of things that are orange or
Ginger stand out to me and I've been
having this idea that we can actually
root that selective
attention to the very deepest level of
our belief
hierarchy such that we kind of do have
it's very Jordan Peterson actually
because it's very sort of narrativity
but we do have these kinds of deep like
what like we have basically a guiding
principle and that could be governed by
your biology by your so by your soou
culture but I really like this idea that
like your value you know I tweeted that
thing that you liked which was Simone
Veil attention is prayer um influenced
by mcil Chris what attention is a moral
act and I think that's really powerful
because attention is basically a
phenomenological indicator of that which
you are Guided
by so I'm thinking maybe again it's a
I'm putting this out there so no one
steals it but I've got this I'm trying
to think about how to formalize this
idea that like the very deep temporary

invariant parts of our being might
actually be the guiding force of our
attention scha that's that's wonderful I
hope you write that paper um I I haven't
done so much research on
attention as such uh in in psychological
literature and such so it's it's it's
really cool to hear that what you've
researched confirms this kind of
perspective well hopefully hopefully
someone doesn't come into the comment
section you know are those comment
sections but people always have things
to say um this might be a slightly
selfish question uh but I don't think it
is because I think people also benefit
from this you say that I wanted to kind
of go back to your
Journey as an academic if you will and
starting off as philosophy and then Andy
Clark and then sort of finding modeling
saying what really helped us actually
modeling a Helm Holtz machine for
example and you hear this a lot you know
if you want to learn how to do pomdp
stuff know how to do the mat lab
work the question I've always had like I
haven't done any of this coding what
does it look like I mean when you say I
code a Helm Holtz machine
what do you get on your screen I know
that sounds really stupid but I think
people are kind of confused about what
that looks like especially when K talks
about oh you get these little agents who
are doing things what what does that
actually look like yeah no I don't it's
not a stupid question it's it's kind of

it's one of the problems you have to
solve if you wanna want to do this stuff
right so so what I did I mean this was
in the relatively early days of this
stuff in academ in broader Academia in
the sense that unless you were doing
machine learning you probably didn't
usually care about this stuff so I did
was I started writing it in JavaScript I
used Dynamic CSS to render this as a web
page you know um I'll have to send I'll
have to send you some link I'll post it
on my Twitter or something yeah
basically I had this very early the
reason I never shared it so much with
the world at the time is that it didn't
work very well at all but I basically
had a Helo machine visualized in a in a
browser and it was like you just have
like um some squares for each and on
there would be some in the hidden layers
you'd have these sort of staticky
looking patterns as you'd expect and
then you'd visualize the output layer as
an image and I sort of overlaid the um
the input the ground truth input image
with the prediction using like red and
blue or something I got those color
codes come from like Jeff hinton's work
and a lot of people have used them um or
red and green maybe but anyway um so
that to me that I really needed this
concrete visualization to like
understand what's going on if you run
Carl's mat lab code it's much more like
you just get the output I mean it's not
it's a that's a fine way of working it's
worked well for for many people right

like you see the belief distribution
change over time um just sort of plotted
on um on an axis like you know uh based
on various conditions um but so yeah so
in in any case I mean you I think you
you need to set up um a visualization
that makes this stuff intelligible I
feel like I don't know if that was the
hard of your question but you no no no
no no
I'm yes I'm just kind of I guess people
always ask like
what what are these like what are you
know when we talk about insilico agents
right is that a thing like really or is
that just a metaphor I
mean I kind of hope it's a metaphor
because I just you know kill thousands
of them in a year if they're real you
know oh yeah yeah yeah yeah yeah sure my
browser window and it's just gone um but
no I mean I think to the extent that you
can model to the extent that you think
these types of models are models of
agency yeah it's real um in your browser
window what what do you actually get you
know what I mean like what what do you
see when you say agent I think people
actually think of little Gummy Bears or
you know little teddy bear looking
things okay well so the helmos machine
then is a bad example because that's
that's really just doing sort of
uh it was really just showing online
learning of like um of a generative
model and it doesn't an action component
to be interesting but um I've seen tons
of visualizations of um of active

inference models and other similar
models so I mean if you look at some of
the um some of the machine learning work
on learning to play video games is a
good example so
so that's a strange case but you know if
you have a a model that's trained to
play Super Mario Brothers then I mean
you see the effects of the agent anyway
um you can you can just visual it
depends on your problem right there's
plenty of um sort of grid World type
simple proof of concept type active
inference models that get made where you
can see the location of an agent on a
little map or something um I love that
yeah but but typically honestly I mean
just as a as a matter of like everyday
sort of um working reality it's like
usually you'll get a series of plots of
the belief belief updates over time and
and the action that was chosen right and
that works yeah and it's an ongoing
struggle to find better more powerful
visualizations for this stuff sure sure
I imagine it is okay sort of just
getting you know wrapping things up
you've gone from philosophy
to computational science let's say um
and the the the kind of contentious
Bridge there let's say between
phenomenal structures and mechanical
explanations or mechanical
formalisms is often called de generative
Passage um started I guess by verel and
then formal or you know that term
instituted by Maxwell and his team in
that 2022

paper I really like that you I think I
think I get this from a lot of
philosophers who start off as
philosophers is the kind of non-
reducibility of intentionality the non-
reducibility of
Consciousness where what do you think of
the generative passage and what do you
think is the sort of most I won't say
likely but the most adaptive or
effective manifestation of it and by
that I mean some people talk about
ontological naturalism which is we could
basically posit as a reductive approach
an epistemological naturalism which is
that the way that we come the way we
think about or describe um let's say in
this case basian Dynamics is going to be
helpful um or actually oh no not in that
case it's not helpful it actually does
describe the structures the intentional
structures or procedural or more you
know mechanical way where it's actually
just useful where it's kind of
instrumental um but actually might not
be telling us anything too concrete
about the phenomenal reality now since
you've kind of flirted in both camps
where uh where do your allegiances lie I
don't mean what do you think yeah no uh
so as far as the narrowly the term
generative passage um I stop me if you
think I'm misusing that term uh I
haven't been deeply involved in thinking
in that term in those terms but I think
you're talking about the relationship
between phenomenology and uh in this
case basian Dynamics let's say or or

um well I mean people talk about
neurophenomenology right so it could be
neuronal Dynamics yeah yeah great so um
yeah so my overall so there's a lot of
things rolled into that question um
epistemologically I kind of see things
as and this it's good that you brought
this up I wanted to comment on this
earlier um so another paper that I
haven't published is on is on this topic
so um like I kind of I would be a
transcendental idealist except that the
except that the evidence that I've seen
for that position is empirical so I kind
of I probably someone said something
similar but I want to I want to write a
paper on that a sort of cont like almost
empirically informed conis where it's
like well what we've learned about how
these systems work
um provides evidence for the conclusion
that the our experience of reality is is
mostly based on prior structures in the
brain right um so I think that's really
interesting epistemological it's
different from sort of classical
transcendental idealism where you just
had K and categories right um because
you're you're sort of presuming to say
something it's there's a sense in which
the the the naturalistically studi
reality comes first there because you're
saying it's like well the way neurons
work we probably en code these
generative models in our brain right and
that's why we conclude that um we have
subjective ontologies and such that we
learn in order to parse out the sensory

data um so so overall that's my that's
my sort of view on the epistemology of
it but um in terms of the relationship
between phenomenology and cognitive
structures and physical structures um I
Tred try to in my work I try to um argue
for the the the most minimal account of
that relationship where where
transparency is the ideal um across all
those levels so of course that's not
going to work perfectly like if I see a
dog on the street doesn't mean there's a
dog in my head it doesn't mean that
something um it does mean that there's
something whose Dynamics reflect the
Dynamics of that thing out there um if
there's a thing out there as we said
earlier um but but let's for a second
let's just think about the relationship
between the phenomenology and the
cognitive structures right um um so yeah
also I cite some work by the gestal
psychologists who really assumed that
there was a strong isomorphism there um
that you really in a sense if you if you
looked closely enough at the um well I
sorry I was I was shifting to the
physical level of description again but
certainly that there you should be able
to sort of read understand the structure
of phenomenology in terms of um in terms
of the structure of these basian prior
prior structures basically and that get
updated by by sensory experience and so
I totally agree with Maxwell at all on
that the one reason I haven't been more
gung-ho about um computational
phenomenology is just that I already

assume something further which is that
like so so their their research program
as I understanded is let sort of step
back and just try to build a generative
model for phenomenology yeah and
understand how you understand
phenomenology in terms of general of
modeling I think you can just shoot
straight through the cognitive structure
and just talk about isomorphism between
the phenomenological structures and the
cognitive structures and the um
implementation bases for those inter so
so I I I guess it's good it's
epistemically humble and it's a good
project to try to do that um I just find
that it's more effective for me to be
constrained by more um the so I try I I
usually when I think about this stuff
I'm constrained by thinking about
neuronal implementation level as well as
cognitive level as well as
phenomenological I was hoping you gon to
be looser oh well I was hoping you got
it's there might be some link um okay
I'm joking you're entitled to of course
you're entitled I mean I
I'm personally I mean I've written about
this a little bit publicly uh
ontologically speaking I'm quite
agnostic um
I have a I have maybe a slightly
romantic um constraint which means that
I just don't want to do the reductionism
I just don't um but actually I should
clarify I've actually got the paper up
in front of me uh Maxwell and colleagues
make it clear that actually the

so-called ontological naturalization
doesn't necessarily entail
reductionism um only so only that and I
can quote here some theal framework
means that one reformulates all the
properties entities and processes
postulated by that theory in terms of
those posited in the ontology of the
empirical
Sciences um and how that gets then
fleshed out in terms of epistemological
naturalization is that you would take
the core principles and forms of your
explanation from the Natural Sciences
and then you would just have
methodological naturalization what I
call procedural naturalization before
which is bringing your sort of
methodology into continuity with the
empirical Sciences um so as you say kind
of constraining
phenomenology uh
you know according to the principles of
Psychology and Neuroscience without
saying that they are those principles or
without saying that we should only
understand them by those principles um
yeah it's a really it's a really nice
trifect to that um and I guess there
isn't an answer but you know you have
some people who are very adamant
that perhaps even the methodological
naturalization is a step too far um and
we should just be as humble as possible
with respect to the nature of cons ious
and say you we just have no idea in what
form it's grounded or in what form it
relates to the physical universe but I I

I'm probably not that loose but it's
inter can I follow up with one more
please um because I I think it's a
really important Point um so I I I don't
think in reductive terms about
Consciousness like so even if even if it
turns out that there's a perfect
isomorphism between phenomenal cognitive
and physical structures
uh I don't think that means that
everything would be reduced to physical
structures like so so I see more I hope
that these lines of inquir would
converge so that you'd end up seeing
that these are different ways of talking
about the same thing but it doesn't mean
that one of those ways is fundamental
right um so yeah I don't know if that is
any different than what you just said
but yeah yeah
yeah yeah no that's absolutely great um
okay final question uh
actually penultimate question we can
rush through these uh pan
psychism is it is it is it or
is it
brilliant
sorry you put it that only because you
mentioned it only because you mentioned
it earlier no no it's certainly not
um I I don't think it's I don't
think it's brilliant to just rush
headlong into it without thinking about
it but um no I think I think it depends
what you mean right but
you assume we're talking about like
phenomenal Consciousness I'm quite
comfortable with the idea that any

physical structure
um there's some edge cases like rocks
where it's like
well you it's kind of like a don't care
case as Quin would say it's like yeah um
it's Ed where there's no there's no
structure to speak of I guess I guess to
me the natural view is that the kinds of
contents you can represent consciously
depend on the complexity of youru
structure but the base ground existence
of sort of a field in which these things
are represented I don't see why that
should be different from like the basis
for physical
existence
good final question yeah you can only
have one for the rest I'm being trivial
now but you can only have one for the
rest of your life philosophy computer
science man which one are you
keeping oh that's rough
um I was going to say computers science
CU I think that the best computer
science contains good philosophy but I
think in terms of the intention rather
than the extension of these things I
think philosophy because you'd end up
inventing computer science if you did
philosophy well enough in the long term
fantastic I'm so glad you said that uh
which which one's
harder um computer science is way harder
is it I mean sorry if hard means
like it's hard to do it's like a
struggle then yes but if you mean like
cognitively
theoretically oh then philosophy

good good yeah yeah we'll have to discuss more about uh about all of this stuff this was so much fun Alex thank you uh thank you for dealing with my Layman thoughts and terrible attempts to trying to understand things that are way Bey on my pay grade but this was awesome likewise thanks Darius this was this was great cool all right thank you everyone for tuning in as well okay